Cuong Than

Ph.D. Candidate in Computer Science | University of Massachusetts Amherst
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Research Interests

Graph algorithms, computational geometry, discrete metric spaces, and theoretical computer science.

Education

University of Massachusetts Amherst, Amherst, MA — 2021–2027 (expected)
Ph.D. in Computer Science; Advisor: Hung Le — *Expected June 2027*

University of Nebraska–Lincoln, Lincoln, NE — 2019–2021
M.S. in Computer Science — *June 2021*

Hanoi University of Science and Technology, Hanoi, Vietnam — 2014–2019
B.S. in Computer Science; highest GPA in the Honors Program — *June 2019*

Honors and Awards

Google PhD Fellowship in Algorithms and Optimization — 2025–2027
- One of three recipients in Algorithms and Optimization; provides two years of research support.

1st Place, ICPC North Central North America Regional Contest — 2019
2nd Prize, ICPC Asia Regional Contest — 2018
2nd Prize, Vietnamese Olympiad in Informatics for Undergraduates — 2018
1st Prize, Vietnamese Olympiad in Analysis for Undergraduates — 2015
1st Prize, Vietnamese Olympiad in Linear Algebra for Undergraduates — 2015

Publications

[14] H. Le, S. Solomon, **C. Than**, C. D. Tóth, and T. Zhang. "Improved Euclidean Shallow-Light Trees." *SODA 2027*, to appear.

[13] A. La, H. Le, S. Solomon, **C. Than**, Vinayak, S. Yang, and T. Zhang. "Optimal Bounds for Spanners and Tree Covers in Doubling Metrics." *SoCG 2026*. Invited to the SoCG 2026 special issue of *Discrete & Computational Geometry*.

[12] H. Le, S. Solomon, L. Milenkovic, and **C. Than**. "Tree-Like Shortcuttings of Trees." *SoCG 2026*. Invited to the SoCG 2026 special issue of *Discrete & Computational Geometry*.

[11] H. Le, S. Solomon, **C. Than**, C. D. Tóth, and T. Zhang. "Approximating Euclidean Shallow-Light Trees." *SoCG 2026*.

[10] H. Le, S. Solomon, **C. Than**, C. D. Tóth, and T. Zhang. "Approximate Light Spanners in Planar Graphs." *SODA 2026*.

[9] H. C. Chang, J. Conroy, H. Le, S. Solomon, and **C. Than**. "Light Tree Covers, Routing, and Path-Reporting Oracles via Spanning Tree Covers in Doubling Graphs." *STOC 2025*.

[8] C. Konrad, A. McGregor, R. Sengupta, and **C. Than**. "Matchings in Low-Arboricity Graphs in the Dynamic Graph Stream Model." *FSTTCS 2024*.

[7] H. Le, S. Solomon, **C. Than**, C. D. Tóth, and T. Zhang. "Towards Instance-Optimal Euclidean Spanners." *FOCS 2024*.

[6] H. C. Chang, J. Conroy, H. Le, L. Milenkovic, S. Solomon, and **C. Than**. "Optimal Euclidean Tree Covers." *Discrete & Computational Geometry*, 75(2):625–659. Conference version: *SoCG 2024*.

[5] H. C. Chang, J. Conroy, H. Le, L. Milenkovic, S. Solomon, and **C. Than**. "Shortcut Partitions in Minor-Free Graphs: Steiner Point Removal, Distance Oracles, Tree Covers, and More." *SODA 2024*.

[4] H. C. Chang, J. Conroy, H. Le, L. Milenkovic, S. Solomon, and **C. Than**. "Covering Planar Metrics (and Beyond): $O(1)$ Trees Suffice." *FOCS 2023*.

Publications (continued)

[3] H. Le, S. Solomon, and **C. Than**. "Optimal Fault-Tolerant Spanners in Euclidean and Doubling Metrics: Breaking the $\Omega(\log n)$ Lightness Barrier." *FOCS 2023*.

[2] H. Le and **C. Than**. "Greedy Spanners in Euclidean Spaces Admit Sublinear Separators." *ACM Transactions on Algorithms (TALG)*. Conference version: *SODA 2022*.

[1] H. Chan, M. Irfan, and **C. Than**. "Schelling Models with Localized Social Influence: A Game-Theoretic Framework." *AAMAS 2020*.

In theoretical computer science, author names are conventionally listed in alphabetical order. Conference abbreviations: STOC — ACM Symposium on Theory of Computing; FOCS — IEEE Symposium on Foundations of Computer Science; SODA — ACM-SIAM Symposium on Discrete Algorithms; SoCG — International Symposium on Computational Geometry; AAMAS — International Conference on Autonomous Agents and Multiagent Systems; FSTTCS — Foundations of Software Technology and Theoretical Computer Science.

Invited Talks

Theory Seminar, Brown University 2026
"Approximating Light Spanners."

Theory Seminar, University of Michigan 2026
"Light Tree Covers, Routing, and Path-Reporting Oracles via Spanning Tree Covers in Doubling Graphs."

Theory Group Seminar, Dartmouth College 2025
"Towards Instance-Optimal Euclidean Spanners."

Conference Presentations

"Approximate Euclidean Shallow-Light Trees," *SoCG* 2026
"Towards Instance-Optimal Euclidean Spanners," *FOCS* 2024
"Optimal Fault-Tolerant Spanners in Euclidean and Doubling Metrics," *FOCS* 2023
"Greedy Spanners in Euclidean Spaces Admit Sublinear Separators," *SODA* 2022

Research Experience

Student Researcher, Google Research Dec. 2024–Mar. 2025
- Developed routing algorithms for temporal graphs with the Google Geo team.

Student Researcher, Google Research Aug.–Oct. 2023
- Worked with the Graph Mining team in the Algorithms and Optimization group.
- Implemented and improved graph-based sampling and distillation algorithms for Gemini.

Research Assistant, University of Massachusetts Amherst 2021–present
- Conduct research in graph algorithms and computational geometry, with emphasis on spanners, tree covers, and metric algorithms.

Visiting Researcher, Tel Aviv University, Israel Aug. 2022
- Research visit supported by the NSF–BSF grant "Towards a Unified Theory of Spanners."
- Conducted research on distributed and fault-tolerant graph algorithms with Shay Solomon.

Mentoring and Professional Service

Research Mentor, Early Research Scholars Program (ERSP), UMass Amherst. Mentored undergraduate students from underrepresented groups in research projects in theoretical computer science.

Reviewer: *Discrete & Computational Geometry*; *SIAM Journal on Discrete Mathematics*; *ACM Journal of Experimental Algorithmics*; STOC 2025–2026; SODA 2022, 2024–2026; FOCS 2024; SoCG 2026; ESA 2026; APPROX 2026; WADS 2019.

Technical Skills

Programming: C++, Python, Java; prior experience with C# and OCaml.
Systems: Familiar with large-scale distributed systems and production research infrastructure.